

# Level 5 mini project-Copy3

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## 1 Mini Project Level 5

### 1.1 Lingling HE

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### 1.3 Summery of the question

Write a game that instruct to search the Boggle monster in the maze of caverns.

### 1.4 Justifications of the program

- I can explain how my program works.
- I wrote this program myself.
- I used Arrays, accessed with a loop.
- I defined and used methods including at least one that is passed and used array arguments.
- I used useful comments for each method.
- I declared some variables within blocks to reduce scope too a minimum.
- I consistently used well chosen variable names that give a clear indication of their use.
- I used final variables for literal constants.

### 1.5 The literate program development

### 1.6 input

### 1.7 what it does

It creates a keyboard. ## Implementation Scanner available to use.

```
[1]: import java.util.Random; // loading the Random library
// input
public static String input(String question)
{
    Scanner s = new Scanner(System.in);
    System.out.println(question);
    return s.nextLine();
}
```

## 1.8 testsing

```
[2]: input("Theory");
```

Theory

1

```
[2]: 1
```

## 1.9 Charactername

### 1.9.1 What it does

This method print the character name that users set. ### Implementation (how it works) It asks for character name as output and keyboard inputs the answer.

```
[3]: public static void charactername()
{
    String name = input("Please type in your character name.");
    System.out.println(name + ". You are in a cavern lit only from the entrance_
    and your task is to capture a Boggle monster.");
    return;
}
```

### 1.10 Testing

```
[4]: charactername ();
```

Please type in your character name.

1

1. You are in a cavern lit only from the entrance and your task is to capture a Boggle monster.

## 2 Virtual dice

### 3 What it does

Creates a program to roll the dice for the amount of starting points the user own. # Implementation Random methods to roll dice from 0-5. The higher score they rolled, the higher points they gain.

```
[5]: //create virtual dice
public static int Randomdice ()
{
    int points = 0;
    Random dice = new Random (); // create a variable
    int d = dice.nextInt(6) + 1;
```

```

    points = d*10;
    System.out.println("The dice is rolled to " + d + ". You get " + points + "
↳points!");
    return points;
}

```

### 3.1 testing

```
[6]: Randomdice();
```

The dice is rolled to 6. You get 60 points!

```
[6]: 60
```

### 3.2 RandomObject

#### 3.2.1 what it does

It randomly show the object ### Implementation It use random method.

```
[7]: // random object
public static String RandomObject()
{
    Random dice = new Random(); // random number genrator.
    String[] objects = {"Magic swords", "Magic ring", "Potion", "Magic carpet"};
    int o = dice.nextInt(objects.length);
    String foundObject = objects[o];
    return foundObject;
} // end of object

```

#### 3.2.2 testing

```
[8]: RandomObject();
```

```
[8]: Magic swords
```

### 3.3 validation

#### 3.3.1 what it does

It checks if input is y/ n ### Implementation if statement and while loop to continue for invalid input.

```
[9]: public static String validation(String input)
{
    while (! input.equalsIgnoreCase("Y") && !input.equalsIgnoreCase("N")){
        System.out.println("Invalid input. Please enter Y or N.");
        input = input("Please enter Y or N");
    }
}

```

```

    }
    return input;
}

```

```
[10]: validation("e");
```

Invalid input. Please enter Y or N.

Please enter Y or N

y

```
[10]: y
```

## 4 multipleActions

### 4.0.1 what it does

It allows players to make decision on what to do next depend on what the have decided. ###  
Implementation Use while loop to repetitively ask players to make decision.

```
[11]: // letting user to decide what they want to do more than one action.
public static String multipleAction (int points, String foundObject)
{
    boolean continuePlay = true;
    String Question = "";
    boolean collectObject = false;
    boolean validCollection = false;

    while (continuePlay){
        String askQuestion = input("What would you like to do next? Enter. 1)␣
↳Searching the room for monsters. 2) Searching for magical objects. 3)␣
↳Leaving the room.");
        if (askQuestion.equals("1")){
            System.out.println("There is no monster in the room.");
        } else if (askQuestion.equals("2")){
            while (!validCollection){
                String decideCollect = input("You found a " + foundObject + ". Would␣
↳you like to collect it? Please enter Y/N.");
                if (decideCollect.equalsIgnoreCase("Y")){
                    collectObject = true;
                    points +=5;
                    System.out.println("You gained 5 points. Now you have " +␣
↳points + " points.");
                    validCollection = true;
                } else if (decideCollect.equalsIgnoreCase("N")){
                    System.out.println("You chose not to collect the " +␣
↳foundObject + ".");
                }
            }
        }
    }
}

```

```

        validCollection =true;
    }
    } // end of validCollection
} else if (askQuestion.equals("3")){
    return "You left the room.";
} else {
    input("Invalid choice. Please enter 1, 2 or 3.");
}

// requests multiple actions if they did not leave the room.
if (continuePlay){
    String ans = "";
    while (!(ans.equalsIgnoreCase("stay") || ans.
↪equalsIgnoreCase("leave"))){
        ans = input("Do you want to continue or leave the room? Please_
↪select from stay/leave");
        if (ans.equalsIgnoreCase("leave")){
            return "You left the room."; // leave the game
        } else if (!ans.equalsIgnoreCase("stay") && ans.
↪equalsIgnoreCase("leave")){
            input("Please selet stay or leave");
        }
    }
}
return Question;
} //end of multipleAction

```

## 4.1 testing

```
[12]: multipleAction(5, "Magic ring");
```

What would you like to do next? Enter. 1) Searching the room for monsters. 2) Searching for magical objects. 3) Leaving the room.

3

```
[12]: You left the room.
```

## 4.2 moveCavern

### 4.2.1 what it does

It givess playyer option to decide whether they want to move between the Caverns. ### Impe- mentation. Using loops to repeat options and if statement to set instructions.

```
[13]: // move between the Cavern.
public static String moveCavern(int points, String foundObject)
```

```

{
    int[] cavernNumbers = {1, 2, 3, 4}; // array of cavern Numbers
    boolean[] Monster = {false, true, false, true}; // monster information
    String[] description = {"This is first Cavern you are in.", "This is second",
↪Cavern.", "This is third Cavern", "This the the last Cavern where you will",
↪fight the final Monster!"};

    final int costPoints = 2; // set the points cost for moving between cavern
    final int lastCavern = 3; // set final cavern number in array
    int currentCavern = 0; // initial cavern number in array
    int gap = 0; // initial the amounts of points less than cost to move.
    String action;

    // ask for decisions until player reaches the last cavern
    while (currentCavern < lastCavern){
        System.out.println(" You are at " + cavernNumbers[currentCavern] + ": " +
↪ description[currentCavern]);

        // detects a monster.
        if (currentCavern == 1 || currentCavern == 3){
            boolean validMonster = false;
            while(!validMonster) {
                String monsterInformation = input ("There is a monster in this",
↪carvern. would you like to fight? Please enter Y or N.");
                if (monsterInformation.equalsIgnoreCase("Y")){
                    validMonster = true;
                    if (foundObject.equals("Magic sword") || foundObject.
↪equals("Magic ring") || foundObject.equals("Potion")){
                        System.out.println("You have a magic object to defeat",
↪the Boggle.");
                        points +=5;
                        System.out.println("Congratulations! You defeated the",
↪monster and gained 5 points. You have " + points + " points now.");
                    } else {
                        points -= 3;
                        System.out.println("You don't have the object required",
↪for defeating the monster. Go to search for it!");
                    }
                } else if (monsterInformation.equalsIgnoreCase("N")){
                    validMonster = true;
                    points -=3;
                    System.out.println("You chose not to fight the monster. You",
↪lose 3 points. You have " + points + " points now.");
                } else{
                    System.out.println("Invalid. Please enter Y or N.");
                }
            }
        }
    }
}

```

```

        } // loop to fight against monster
    } // end of monster

    // final cavern
    if (currentCavern == lastCavern){
        System.out.println("You are at the final Cavern.");

        // check if player has object to defeat Boggle..
        if (foundObject.equals("Magic sword") || foundObject.equals("Magic_
↪ring") || foundObject.equals("Potion")){
            System.out.println("You have a magic objet to defeat the Boggle!
↪");

            // check whether player has enough points to defeat boggle.
            if( points >= 10){
                System.out.println("You are powerful to defeat the Boggle!
↪");

                } else{
                    gap = 10-points;
                    System.out.println("You need " + gap + " more points to_
↪defeat the Boggle");
                }
            } else {
                System.out.println("You need to collect an object to defeat the_
↪Boggle");
            }
        }

        // ask player whether moving to next cavern.
        String Movedecision = input("You are currently at Cavern " +_
↪cavernNumbers[currentCavern] + ". Do you want to move to next Cavern? Enter Y/_
↪N.");

        // Y responses.
        if (Movedecision.equalsIgnoreCase("Y")){
            // Y response validation.
            if(points >= costPoints){
                points -= costPoints;
                currentCavern ++; // move to next cavern.
                System.out.println("You moved to Cavern " +_
↪cavernNumbers[currentCavern] + ". You have " + points + " points left.");
                System.out.println("Cavern " + cavernNumbers[currentCavern] + ":
↪ " + description[currentCavern]);

                // call the multipleAction method. multiple actions the player_
↪decide to do.
                action = multipleAction(points, foundObject);
                if(action.equals("leave")){

```

```

        return "The end of the game."; // end game if they left
↳the room.
    }
    } else{
        System.out.println("You don't have enough points. Go and search
↳for objects."); // stop moving if they dont have enough points.
        action = multipleAction(points, foundObject); // recall
↳multipleAction if player does not have enough point.
        if (action.equalsIgnoreCase("leave")){
            return "The end of the game.";
        }
    }
    //respond N: stay
    } else if (Movedecision.equalsIgnoreCase("N")){
        System.out.println("You stayed in Cavern " +
↳cavernNumbers[currentCavern]);
        // call multipleAction
        action = multipleAction(points, foundObject);
        if (action.equalsIgnoreCase("leave")){
            System.out.println("Thank you for playing the game. You left
↳the cavern.");
            return "The end of the game"; // exit the game
        }
    } else {
        input("Invalid option. Please enter Y or N."); // validation
    }
}
return "The end of the game"; // end the game.
} // end of moveCavern

```

#### 4.2.2 testing

```
[14]: moveCavern(5, "Magic ring");
```

You are at 1: This is first Cavern you are in.  
 You are currently at Cavern 1. Do you want to move to next Cavern? Enter Y/N.

y

You moved to Cavern 2. You have 3 points left.

Cavern 2: This is second Cavern.

What would you like to do next? Enter. 1) Searching the room for monsters. 2)  
 Searching for magical objects. 3) Leaving the room.

2

You found a Magic ring. Would you like to collect it? Please enter Y/N.

n



You chose not to collect the Magic ring.

Do you want to continue or leave the room? Please select from stay/leave

leave

You are at 2: This is second Cavern.

There is a monster in this cavern. would you like to fight? Please enter Y or N.

y

You have a magic object to defeat the Boggle.

Congratulations! You defeated the monster and gained 5 points. You have 8 points now.

You are currently at Cavern 2. Do you want to move to next Cavern? Enter Y/N.

y

You moved to Cavern 3. You have 6 points left.

Cavern 3: This is third Cavern

What would you like to do next? Enter. 1) Searching the room for monsters. 2) Searching for magical objects. 3) Leaving the room.

3

You are at 3: This is third Cavern

You are currently at Cavern 3. Do you want to move to next Cavern? Enter Y/N.

y

You moved to Cavern 4. You have 4 points left.

Cavern 4: This the the last Cavern where you will fight the final Monster!

What would you like to do next? Enter. 1) Searching the room for monsters. 2) Searching for magical objects. 3) Leaving the room.

3

[14]: The end of the game

### 4.3 The complete program

```
[16]: import java.util.*; // loading the library
//start of BoggleMonster
class BoggleMonster
{
    public static void main(String [] a)//Start of main
    {
        charactername();
        System.out.println("There is a silver ring of invisibility on a
↪ stalagmite in the middle of the carven");
        int points = Randomdice();
        String foundObject = RandomObject();
```

```

        multipleAction(points, foundObject);
        moveCavern(points, foundObject);
        return;
    } //end main

    // input
    public static String input(String question)
    {
        Scanner s = new Scanner(System.in);
        System.out.println(question);
        return s.nextLine();
    }

    //Ask for character name
    public static String charactername()
    {
        String name = input("Please type in your character name.");
        System.out.println(name + ". You are in a cavern lit only from the␣
↪ entrance and your task is to capture a Boggle monster.");
        return name;
    } //end of charactername

    //create virtual dice
    public static int Randomdice ()
    {
        int points = 0;
        Random dice = new Random (); // create a variable
        int d = dice.nextInt(6) + 1;
        points = d*10;
        System.out.println("The dice is rolled to " + d + ". You get " + points␣
↪ + " points!");
        return points;
    }

    // random object
    public static String RandomObject()
    {
        Random dice = new Random(); // random number genrator.
        String[] objects = {"Magic swords", "Magic ring", "Potion", "Magic␣
↪ carpet"};
        int o = dice.nextInt(objects.length);
        String foundObject = objects[o];
        return foundObject;
    } // end of object

    public static String validation(String input)
    {

```

```

        while (! input.equalsIgnoreCase("Y") && !input.equalsIgnoreCase("N")){
            System.out.println("Invalid input. Please enter Y or N.");
            input = input("Please enter Y or N");
        }
        return input;
    }

    // letting user to decide what they want to do more than one action.
    public static String multipleAction (int points, String foundObject)
    {
        boolean continuePlay = true;
        String Question = "";
        boolean collectObject = false;
        boolean validCollection = false;

        while (continuePlay){
            String askQuestion = input("What would you like to do next? Enter.␣
↪1) Searching the room for monsters. 2) Searching for magical objects. 3)␣
↪Leaving the room.");
            if (askQuestion.equals("1")){
                System.out.println("There is no monster in the room.");
            } else if (askQuestion.equals("2")){
                while (!validCollection){
                    String decideCollect = input("You found a " + foundObject␣
↪+ ". Would you like to collect it? Please enter Y/N.");
                    if (decideCollect.equalsIgnoreCase("Y")){
                        collectObject = true;
                        points +=5;
                        System.out.println("You gained 5 points. Now you have "␣
↪+ points + " points.");
                        validCollection = true;
                    } else if (decideCollect.equalsIgnoreCase("N")){
                        System.out.println("You chose not to collect the " +␣
↪foundObject + ".");
                        validCollection =true;
                    }
                } // end of validCollection
            } else if (askQuestion.equals("3")){
                return "You left the room.";
            } else {
                input("Invalid choice. Please enter 1, 2 or 3.");
            }

            // requests multiple actions if they did not leave the room.
            if (continuePlay){
                String ans = "";

```

```

        while (!(ans.equalsIgnoreCase("stay") || ans.
↪equalsIgnoreCase("leave"))){
            ans = input("Do you want to continue or leave the room?_
↪Please select from stay/leave");
            if (ans.equalsIgnoreCase("leave")){
                return "You left the room."; // leave the game
            } else if (!(ans.equalsIgnoreCase("stay") && ans.
↪equalsIgnoreCase("leave")){
                input("Please selet stay or leave");
            }
        }
    }
    return Question;
} //end of multipleAction

// move between the Cavern.
public static String moveCavern(int points, String foundObject)
{
    int[] cavernNumbers = {1, 2, 3, 4}; // array of cavern Numbers
    boolean[] Monster = {false, true, false, true}; // monster information
    String[] description = {"This is first Cavern you are in.", "This is_
↪second Cavern.", "This is third Cavern", "This the the last Cavern where you_
↪will fight the final Monster!"};

    final int costPoints = 2; // set the points cost for moving between_
↪cavern

    final int lastCavern = 3; // set final cavern number in array
    int currentCavern = 0; // initial cavern number in array
    int gap = 0; // initial the amounts of points less than cost to move.
    String action;

    // ask for decisions until player reaches the last cavern
    while (currentCavern < lastCavern){
        System.out.println(" You are at " + cavernNumbers[currentCavern] +_
↪": " + description[currentCavern]);

        // detects a monster.
        if (currentCavern == 1 || currentCavern == 3){
            boolean validMonster = false;
            while(!validMonster) {
                String monsterInformation = input ("There is a monster in this_
↪carvern. would you like to fight? Please enter Y or N.");
                if (monsterInformation.equalsIgnoreCase("Y")){
                    validMonster = true;

```

```

        if (foundObject.equals("Magic sword") || foundObject.
↪equals("Magic ring") || foundObject.equals("Potion")){
            System.out.println("You have a magic object to defeat
↪the Boggle.");

            points +=5;
            System.out.println("Congratulations! You defeated the
↪monster and gained 5 points. You have " + points + " points now.");
        } else {
            points -= 3;
            System.out.println("You don't have the object required
↪for defeating the monster. Go to search for it!");
        }
    } else if (monsterInformation.equalsIgnoreCase("N")){
        validMonster = true;
        points -=3;
        System.out.println("You chose not to fight the monster. You
↪lose 3 points. You have " + points + " points now.");
    } else{
        System.out.println("Invalid. Please enter Y or N.");
    }
} // loop to fight against monster
} // end of monster

// final cavern
if (currentCavern == lastCavern){
    System.out.println("You are at the final Cavern.");

    // check if player has object to defeat Boggle..
    if (foundObject.equals("Magic sword") || foundObject.equals("Magic
↪ring") || foundObject.equals("Potion")){
        System.out.println("You have a magic objet to defeat the Boggle!
↪");

        // check whether player has enough points to defeat boggle.
        if( points >= 10){
            System.out.println("You are powerful to defeat the Boggle!
↪");
        } else{
            gap = 10-points;
            System.out.println("You need " + gap + " more points to
↪defeat the Boggle");
        }
    } else {
        System.out.println("You need to collect an object to defeat the
↪Boggle");
    }
}
}

```

```

        // ask player whether moving to next cavern.
        String Movedecision = input("You are currently at Cavern " +
↪cavernNumbers[currentCavern] + ". Do you want to move to next Cavern? Enter Y/
↪N.");

        // Y responses.
        if (Movedecision.equalsIgnoreCase("Y")){
            // Y response validation.
            if(points >= costPoints){
                points -= costPoints;
                currentCavern ++; // move to next cavern.
                System.out.println("You moved to Cavern " +
↪cavernNumbers[currentCavern] + ". You have " + points + " points left.");
                System.out.println("Cavern " + cavernNumbers[currentCavern] + ":
↪ " + description[currentCavern]);

                // call the multipleAction method. multiple actions the player
↪decide to do.
                action = multipleAction(points, foundObject);
                if(action.equals("leave")){
                    return "The end of the game."; // end game if they left
↪the room.
                }
            } else{
                System.out.println("You don't have enough points. Go and search
↪for objects."); // stop moving if they dont have enough points.
                action = multipleAction(points, foundObject); // recall
↪multipleAction if player does not have enough point.
                if (action.equalsIgnoreCase("leave")){
                    return "The end of the game.";
                }
            }
            //respond N: stay
        } else if (Movedecision.equalsIgnoreCase("N")){
            System.out.println("You stayed in Cavern " +
↪cavernNumbers[currentCavern]);
            // call multipleAction
            action = multipleAction(points, foundObject);
            if (action.equalsIgnoreCase("leave")){
                System.out.println("Thank you for playing the game. You left
↪the cavern.");
                return "The end of the game"; // exit the game
            }
        } else {
            input("Invalid option. Please enter Y or N."); // validation
        }
    }
}

```

```
    }  
    return "The end of the game"; // end the game.  
    } // end of moveCavern  
} //end of class
```

```
[ ]: BoggleMonster.main( null)
```

Please type in your character name.

qfwef

qfwef. You are in a cavern lit only from the entrance and your task is to capture a Boggle monster.

There is a silver ring of invisibility on a stalagmite in the middle of the carven

The dice is rolled to 4. You get 40 points!

What would you like to do next? Enter. 1) Searching the room for monsters. 2) Searching for magical objects. 3) Leaving the room.

1

There is no monster in the room.

Do you want to continue or leave the room? Please select from stay/leave

stay

What would you like to do next? Enter. 1) Searching the room for monsters. 2) Searching for magical objects. 3) Leaving the room.

2

You found a Magic carpet. Would you like to collect it? Please enter Y/N.

n

You chose not to collect the Magic carpet.

Do you want to continue or leave the room? Please select from stay/leave

leave

You are at 1: This is first Cavern you are in.

You are currently at Cavern 1. Do you want to move to next Cavern? Enter Y/N.

n

You stayed in Cavern 1

What would you like to do next? Enter. 1) Searching the room for monsters. 2) Searching for magical objects. 3) Leaving the room.

3

You are at 1: This is first Cavern you are in.

You are currently at Cavern 1. Do you want to move to next Cavern? Enter Y/N.

y

You moved to Cavern 2. You have 38 points left.

Cavern 2: This is second Cavern.

What would you like to do next? Enter. 1) Searching the room for monsters. 2) Searching for magical objects. 3) Leaving the room.

3

You are at 2: This is second Cavern.

There is a monster in this cavern. would you like to fight? Please enter Y or N.

y

You don't have the object required for defeating the monster. Go to search for it!

You are currently at Cavern 2. Do you want to move to next Cavern? Enter Y/N.

n

You stayed in Cavern 2

What would you like to do next? Enter. 1) Searching the room for monsters. 2) Searching for magical objects. 3) Leaving the room.

2

You found a Magic carpet. Would you like to collect it? Please enter Y/N.

y

You gained 5 points. Now you have 40 points.

Do you want to continue or leave the room? Please select from stay/leave

leave

You are at 2: This is second Cavern.

There is a monster in this cavern. would you like to fight? Please enter Y or N.

y

You don't have the object required for defeating the monster. Go to search for it!

You are currently at Cavern 2. Do you want to move to next Cavern? Enter Y/N.

n

You stayed in Cavern 2

What would you like to do next? Enter. 1) Searching the room for monsters. 2) Searching for magical objects. 3) Leaving the room.

2

You found a Magic carpet. Would you like to collect it? Please enter Y/N.

y

You gained 5 points. Now you have 37 points.

Do you want to continue or leave the room? Please select from stay/leave



leave

You are at 2: This is second Cavern.

There is a monster in this cavern. would you like to fight? Please enter Y or N.

y

You don't have the object required for defeating the monster. Go to search for it!

You are currently at Cavern 2. Do you want to move to next Cavern? Enter Y/N.

n

You stayed in Cavern 2

What would you like to do next? Enter. 1) Searching the room for monsters. 2) Searching for magical objects. 3) Leaving the room.

1

There is no monster in the room.

Do you want to continue or leave the room? Please select from stay/leave

leave

You are at 2: This is second Cavern.

There is a monster in this cavern. would you like to fight? Please enter Y or N.

y

You don't have the object required for defeating the monster. Go to search for it!

You are currently at Cavern 2. Do you want to move to next Cavern? Enter Y/N.

y

You moved to Cavern 3. You have 24 points left.

Cavern 3: This is third Cavern

What would you like to do next? Enter. 1) Searching the room for monsters. 2) Searching for magical objects. 3) Leaving the room.

2

You found a Magic carpet. Would you like to collect it? Please enter Y/N.

y

You gained 5 points. Now you have 29 points.

Do you want to continue or leave the room? Please select from stay/leave

stay

What would you like to do next? Enter. 1) Searching the room for monsters. 2) Searching for magical objects. 3) Leaving the room.

2

Do you want to continue or leave the room? Please select from stay/leave  
leave

You are at 3: This is third Cavern  
You are currently at Cavern 3. Do you want to move to next Cavern? Enter Y/N.

y

You moved to Cavern 4. You have 22 points left.

Cavern 4: This the the last Cavern where you will fight the final Monster!

What would you like to do next? Enter. 1) Searching the room for monsters. 2)  
Searching for magical objects. 3) Leaving the room.

[ ]: